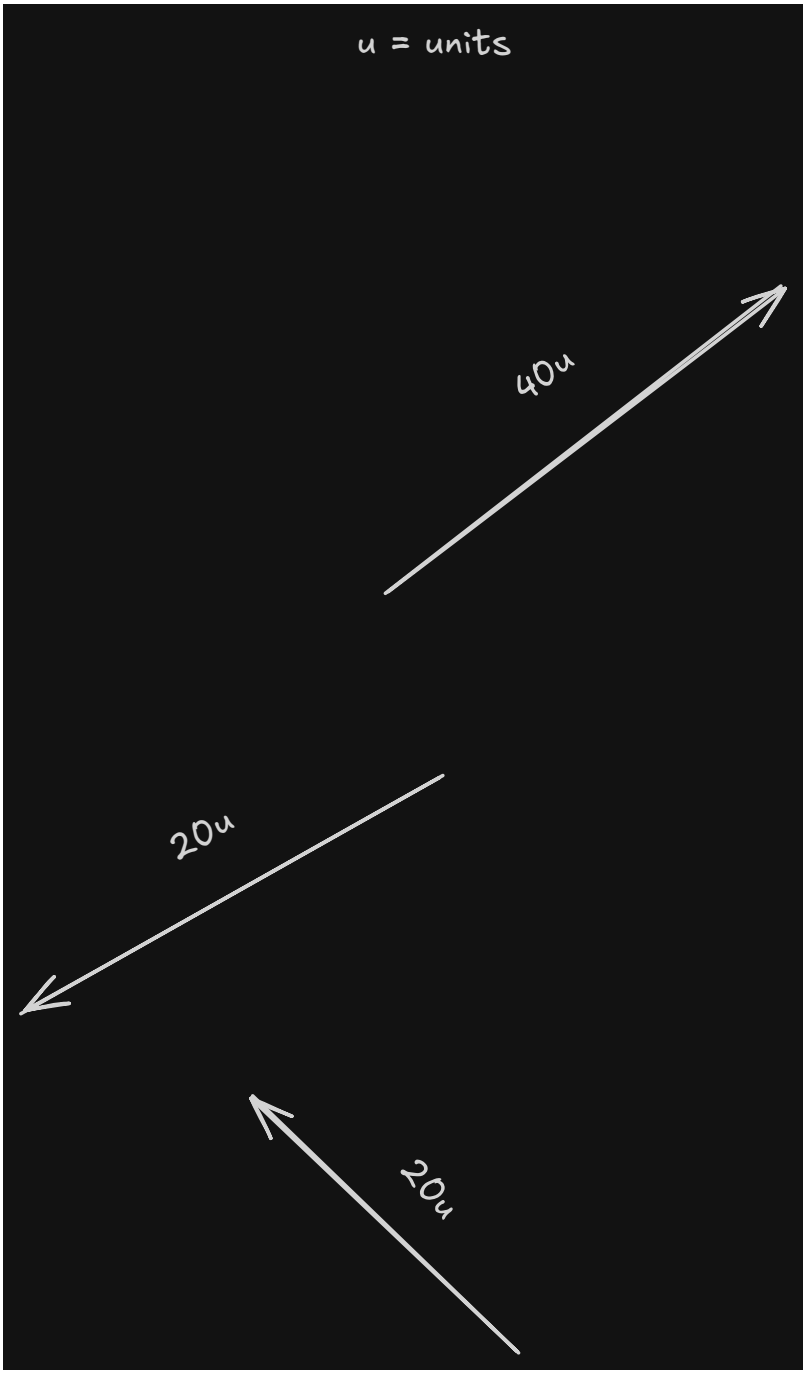
$|\vec{v}|$ is the magnitude
θ is the angle
v_x is the x position of the vector
v_y is the y position of the vector
the direction and position are different for 3
the direction, position, and angle are different for 4



No because those are the most basic components of a vector

they are all $10u$
a has a negative angle
b is northeast and c is south
yes assuming both runners start at the same place

because they have the same magnitude and angle

I didn't get to the end but I learned a lot about Vectors